

# Geodesic Momentum Sampling for Masked Discrete Diffusion Language Models

An Inference-Time Fisher–Rao Extrapolation Method

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## Abstract

Masked discrete diffusion language models generate text through a sequence of categorical denoising transitions. We study an inference-time modification that treats each per-token categorical prediction as a point on the Fisher–Rao statistical manifold. The square-root map embeds the categorical simplex into the positive orthant of a hypersphere, where closed-form spherical interpolation is available. At each reverse step, we recursively continue the great-circle arc from the previous extrapolated prediction through the current model prediction, map the result back to a categorical distribution, and insert it into the original MDLM or SEDD transition. We call the method *geodesic momentum sampling*.

We give a precise geometric derivation, prove equivalence between the unprojected update and an exponential-map continuation, and show that its small-angle coefficients coincide with the coefficients of variable-step Adams–Bashforth two-step extrapolation. We also delimit this analogy: the method extrapolates prediction distributions rather than tangent vector fields, retains stochastic categorical jumps, and uses a positive-orthant projection. It is therefore not a second-order probability-flow ODE solver, and no second-order convergence claim is made. Archived experiments on OpenWebText checkpoints suggest improved GPT-2 generative perplexity for MDLM at 10–100 reverse steps and for SEDD at 10–50 steps.

## 1 Introduction

Discrete diffusion language models offer parallel, bidirectional generation, but accurate sampling often requires many reverse transitions. MDLM [Sahoo et al., 2024] simplifies absorbing-state diffusion through a substitution parameterization and provides an efficient cached sampler. SEDD [Lou et al., 2024] instead learns concrete score ratios and admits an analytic reverse transition. Both models nevertheless operate by repeatedly producing categorical distributions over a vocabulary.

Information geometry supplies a natural geometry for these distributions. The interior of the categorical simplex carries the Fisher–Rao metric [Amari, 2016]; the square-root map identifies it, up to a constant scale, with the positive orthant of a sphere. This construction underlies Fisher-Flow [Davis et al., 2024] and the continuous Riemannian diffusion language model (RDLM) [Jo and Hwang, 2025]. Those methods learn or simulate continuous flows on the sphere. Our setting is different: we preserve pretrained discrete MDLM and SEDD models and modify only their inference-time categorical predictions.

The main contributions are:

- We formulate recursive geodesic extrapolation of categorical predictions using the exact square-root embedding and spherical linear interpolation (SLERP).
- We show that the unprojected update is an exponential-map continuation and derive its local Adams–Bashforth-like coefficient structure.
- We instantiate the same geometric operator at two different interfaces: the clean-token prediction in MDLM and the analytic transition distribution in SEDD.

- We state invariants and the precise limits of the high-order analogy. In particular, we do not identify the algorithm with an RDLM probability-flow ODE solver.
- We report MDLM and SEDD quality results across six reverse-step counts.

## 2 Background

### 2.1 Categorical Fisher–Rao geometry

Let

$$\Delta^{V-1} = \left\{ p \in \mathbb{R}^V : p_k \geq 0, \sum_{k=1}^V p_k = 1 \right\} \quad (1)$$

be the categorical probability simplex. On its interior, a tangent vector satisfies  $\sum_k v_k = 0$ , and the Fisher–Rao metric is

$$g_p(v, w) = \sum_{k=1}^V \frac{v_k w_k}{p_k}. \quad (2)$$

**Proposition 1** (Square-root isometry). *Define  $\Psi : \text{int}(\Delta^{V-1}) \rightarrow \mathbb{S}_+^{V-1}(2)$  by  $\Psi(p) = 2\sqrt{p}$  elementwise, where  $\mathbb{S}_+^{V-1}(2)$  is the positive orthant of the sphere of radius two. Then  $\Psi$  is an isometry between the Fisher–Rao metric and the metric induced by the ambient Euclidean space.*

*Proof.* For  $\psi_k = 2\sqrt{p_k}$ , we have  $d\psi_k = dp_k/\sqrt{p_k}$ . Therefore

$$\|d\psi\|_2^2 = \sum_k \frac{dp_k^2}{p_k} = ds_{\text{FR}}^2. \quad (3)$$

Moreover,  $\|\psi\|_2^2 = 4\sum_k p_k = 4$ , so  $\psi$  lies on the radius-two sphere.  $\square$

The implementation uses the unit-sphere coordinate

$$u(p) = \sqrt{p} \in \mathbb{S}_+^{V-1}, \quad (4)$$

which differs only by the constant factor two. For  $p, q \in \text{int}(\Delta^{V-1})$ , their spherical angle is

$$\omega(p, q) = \arccos\langle u(p), u(q) \rangle = \arccos\left(\sum_k \sqrt{p_k q_k}\right), \quad (5)$$

and the Fisher–Rao distance under the radius-two convention is  $d_{\text{FR}}(p, q) = 2\omega(p, q)$ . This formula extends continuously to the closed simplex and yields its metric completion; boundary points themselves are not smooth points of the Fisher–Rao Riemannian manifold.

### 2.2 Masked discrete diffusion and MDLM

Let  $\mathbf{m}$  denote an absorbing mask state and let  $\sigma(t)$  be the integrated noise schedule. The probability of having moved to the mask by time  $t$  is

$$m(t) = 1 - \exp[-\sigma(t)]. \quad (6)$$

For a reverse step  $t > s$ , MDLM predicts a clean-token distribution  $p_{\theta,t}(\cdot | x_t)$  whose absorbing-state parameterization satisfies  $p_{\theta,t}(\mathbf{m} | x_t) = 0$  and  $\sum_{k \neq \mathbf{m}} p_{\theta,t}(k | x_t) = 1$ . When  $x_t = \mathbf{m}$ , the baseline implementation samples from the unnormalized weights

$$\hat{q}_{s|t}^{\text{MDLM}}(k | x_t = \mathbf{m}) = \begin{cases} [m(t) - m(s)]p_{\theta,t}(k | x_t), & k \neq \mathbf{m}, \\ m(s), & k = \mathbf{m}. \end{cases} \quad (7)$$

Their common normalization is immaterial for categorical sampling. When  $x_t \neq \mathfrak{m}$ , the absorbing reverse sampler copies the already unmasked token. The cached sampler reuses  $p_{\theta,t}$  until the discrete state changes [Sahoo et al., 2024].

### 2.3 SEDD analytic transitions

SEDD learns a concrete score ratio that approximates  $p_t(y)/p_t(x)$  [Lou et al., 2024]. Let  $r_\theta(x_t, t)$  denote the learned score vector, let  $\mathcal{S}_{\Delta\sigma}$  denote the staggered-score transform, and let  $K_{\Delta\sigma}(x_t, \cdot)$  denote the transported forward kernel. The analytic sampler uses

$$\hat{q}_{s|t}^{\text{SEDD}} = \mathcal{S}_{\Delta\sigma}(r_\theta(x_t, t)) \odot K_{\Delta\sigma}(x_t, \cdot), \quad \Delta\sigma = \sigma(t) - \sigma(s), \quad (8)$$

followed by categorical sampling. Our SEDD variant applies geometry to the normalized version of equation (8), not to the score vector itself.

## 3 Geodesic Momentum Sampling

### 3.1 A common distribution interface

For each batch element and sequence position, define a categorical distribution  $z_i \in \Delta^{V-1}$  at reverse index  $i$ :

$$z_i = \begin{cases} p_{\theta,t_i}(\cdot | x_{t_i}), & \text{MDLM,} \\ \text{Normalize}(\hat{q}_{t_{i+1}|t_i}^{\text{SEDD}}), & \text{SEDD.} \end{cases} \quad (9)$$

The raw spherical coordinate is  $u_i = \sqrt{z_i}$ . Let  $\hat{u}_{i-1}$  be the previous *extrapolated* coordinate. This recursive choice is important: the method has persistent momentum rather than retaining only two raw predictions.

### 3.2 Exact spherical continuation

For unit vectors  $a, b \in \mathbb{S}^{V-1}$  with angle  $\omega = \arccos\langle a, b \rangle \in (0, \pi)$ , SLERP is [Shoemake, 1985]

$$\text{Slerp}(a, b; \tau) = \frac{\sin[(1-\tau)\omega]}{\sin \omega} a + \frac{\sin(\tau\omega)}{\sin \omega} b. \quad (10)$$

Interpolation uses  $\tau \in [0, 1]$ . We instead choose  $\tau = 1 + \alpha_i$ :

$$e_i = \frac{\sin(-\alpha_i\omega_i)}{\sin \omega_i} \hat{u}_{i-1} + \frac{\sin[(1+\alpha_i)\omega_i]}{\sin \omega_i} u_i, \quad \omega_i = \arccos\langle \hat{u}_{i-1}, u_i \rangle. \quad (11)$$

**Proposition 2** (Exponential-map representation). *For distinct, non-antipodal  $a, b \in \mathbb{S}^{V-1}$  and  $\alpha \geq 0$ ,*

$$\text{Slerp}(a, b; 1 + \alpha) = \text{Exp}_b[-\alpha \text{Log}_b(a)]. \quad (12)$$

Thus equation (11) exactly continues the great-circle arc from  $a$  through  $b$  by an additional fraction  $\alpha$  before the positivity operation described below.

*Proof.* The curve  $\gamma(\tau) = \text{Slerp}(a, b; \tau)$  is the unique shortest great-circle segment from  $a$  to  $b$  for  $\tau \in [0, 1]$ . At  $b = \gamma(1)$ ,  $-\text{Log}_b(a)$  is the forward continuation tangent with norm  $\omega$ . Applying  $\text{Exp}_b$  to the scaled vector  $-\alpha \text{Log}_b(a)$  advances by arc length  $\alpha\omega$ , which is exactly  $\gamma(1 + \alpha)$ .  $\square$

At  $a = b$  (equivalently,  $\omega = 0$ ), we define  $\text{Slerp}(a, a; \tau) = a$  by continuous extension, and the proposition remains valid. Equations (10)–(12) describe the ideal, unclamped operator. For numerical stability, the implementation evaluates the angle as  $\arccos(\text{clamp}(\langle a, b \rangle, -1 + \varepsilon, 1 - \varepsilon))$ ; hence the exact exponential-map

identity applies to the ideal operator, while the implemented rule is a stabilized approximation near the endpoints.

### 3.3 Positive-orthant projection and probability recovery

Great-circle extrapolation may leave the positive orthant. The archived implementation uses

$$\text{Proj}_+(e) = \frac{[e]_+}{\|[e]_+\|_2}, \quad [e]_{+,k} = \max(e_k, 0), \quad (13)$$

and recovers

$$\hat{z}_{i,k} = (\hat{u}_{i,k})^2, \quad \hat{u}_i = \text{Proj}_+(e_i). \quad (14)$$

**Proposition 3** (Probability validity). *Whenever  $\|[e]_+\|_2 > 0$ , the recovered vector  $\hat{z}_i$  lies in  $\Delta^{V-1}$ .*

*Proof.* Each component is a square and is therefore nonnegative. Since  $\hat{u}_i$  has unit norm,  $\sum_k \hat{z}_{i,k} = \sum_k \hat{u}_{i,k}^2 = 1$ .  $\square$

This projection is a practical domain correction, not part of the exact Fisher–Rao geodesic. It is nonsmooth where components cross zero (and undefined when  $[e]_+ = 0$ ), preventing direct inheritance of a high-order geodesic-integrator guarantee.

### 3.4 Step coefficient and the AB2 analogy

Let the descending reverse grid be  $t_0 > t_1 > \dots > t_N$  and define positive step magnitudes  $h_i = t_i - t_{i+1}$ . The implementation uses

$$\alpha_i = c_i \frac{h_i}{2h_{i-1}}, \quad \alpha_0 = 0. \quad (15)$$

For SEDD, the archived runs use  $c_i = 1$ . For MDLM,

$$c_i = c \min\left(1, \frac{h_i}{h_{\text{ref}}}\right), \quad c = 1, \quad h_{\text{ref}} = \frac{1}{50}. \quad (16)$$

**Proposition 4** (Local linear form). *For fixed  $\alpha$  and  $\omega = \arccos\langle a, b \rangle \rightarrow 0$ ,*

$$\text{Slerp}(a, b; 1 + \alpha) = (1 + \alpha)b - \alpha a + O(\omega^2). \quad (17)$$

*With  $c_i = 1$  and  $r_i = h_i/h_{i-1}$ , the leading coefficients are  $(1 + r_i/2, -r_i/2)$ , matching the coefficients of variable-step AB2.*

*Proof.* Using  $\sin(\beta\omega)/\sin\omega = \beta + O(\omega^2)$  in equation (10) with  $\tau = 1 + \alpha$  gives  $\sin(-\alpha\omega)/\sin\omega = -\alpha + O(\omega^2)$  and  $\sin[(1 + \alpha)\omega]/\sin\omega = 1 + \alpha + O(\omega^2)$ .  $\square$

**Remark 1** (Why this is not Riemannian AB2). *Riemannian AB2 combines vector fields  $V_i \in T_{X_i}\mathcal{M}$  after parallel-transporting  $V_{i-1}$  into  $T_{X_i}\mathcal{M}$ , then updates the state with an exponential map [Hairer et al., 2006]. Equation (17) combines points representing categorical predictions; it neither evaluates nor transports tangent vector fields. Coefficient agreement is therefore a local analogy, not an order proof.*

### 3.5 Instantiation for MDLM and SEDD

For MDLM,  $\hat{z}_i$  replaces  $p_{\theta, t_i}$  in equation (7); already unmasked tokens remain fixed. For SEDD,  $\hat{z}_i$  directly replaces the normalized analytic transition in equation (8). In both cases, the next token state is sampled with Gumbel-max. The algorithm therefore keeps all neural-network inputs discrete.

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**Algorithm 1** Geodesic momentum sampling for MDLM or SEDD

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**Require:** pretrained model, reverse grid  $t_0 > \dots > t_N$ , mode  $\in \{\text{MDLM}, \text{SEDD}\}$

```
1:  $x \leftarrow$  all-mask prior;  $\hat{u}_{-1} \leftarrow \emptyset$ 
2: for  $i = 0, \dots, N - 1$  do
3:   if mode is MDLM then
4:      $z_i \leftarrow p_{\theta, t_i}(\cdot \mid x)$ 
5:   else
6:     form  $\tilde{q}_i$  by equation (8);  $z_i \leftarrow \text{Normalize}(\tilde{q}_i)$ 
7:    $u_i \leftarrow \sqrt{z_i}$ ; compute  $\alpha_i$  by equation (15)
8:   if  $i = 0$  then
9:      $\hat{u}_i \leftarrow u_i$ 
10:  else
11:     $e_i \leftarrow \text{Slerp}(\hat{u}_{i-1}, u_i; 1 + \alpha_i)$ 
12:     $\hat{u}_i \leftarrow \text{Proj}_+(e_i)$ 
13:   $\hat{z}_i \leftarrow \hat{u}_i^{\odot 2}$ 
14:  if mode is MDLM then
15:    insert  $\hat{z}_i$  into equation (7); sample masked positions and copy others
16:  else
17:     $x \sim \text{Cat}(\hat{z}_i)$ 
18: apply the baseline final denoising step and return  $x$ 
```

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## 4 Theoretical Scope and Non-Claims

The method preserves several exact properties:

1. **Discrete-state compatibility.** The model always receives integer token indices.
2. **First-step equivalence.** Since  $\alpha_0 = 0$ , the first categorical transition equals the corresponding baseline transition under the same random draw.
3. **Simplex validity.** Subject to the nonzero projection condition, every extrapolated distribution is nonnegative and normalized.
4. **MDLM absorbing structure.** Tokens already unmasked remain fixed, exactly as in the uncached MDLM reverse transition.

However, the sampled sequence  $x_{t_i}$  changes stochastically, so the sequence  $z_i$  is a collection of conditional predictions along a random path, not a deterministic marginal curve  $p_t$ . In SEDD,  $z_i$  is a one-step transition distribution rather than a state of a probability-flow ODE. Moreover, the Fisher–Rao metric is singular on the boundary  $p_k = 0$ , while MDLM intentionally sets the mask probability to zero and produces delta distributions for already unmasked tokens. The square-root representation remains numerically usable there, but smooth-manifold Taylor assumptions do not.

RDLM constructs a continuous bridge-mixture SDE on the hypersphere and trains a model conditioned on continuous states [Jo and Hwang, 2025]; Fisher-Flow learns a continuous vector field transporting probability mass along spherical geodesics [Davis et al., 2024]. Our method does neither. Likewise, DPM-Solver obtains high order by exploiting an exact semilinear integral form of a diffusion ODE [Lu et al., 2022], a structure absent from the present discrete sampler. We therefore make no claim of marginal preservation, local truncation error  $O(h^3)$ , or global convergence  $O(h^2)$ .

## 5 Experimental Evaluation

### 5.1 Protocol

The scripts compare MDLM against its `ddpm_cache` sampler and SEDD against its analytic sampler. Both use OpenWebText checkpoints in the MDLM codebase and sequence length 1024. The reverse-step counts are 10,

**Table 1:** Quality results.  $\text{Gain}_{\text{PPL}} = 100(\text{baseline} - \text{geodesic})/\text{baseline}$ , so positive values favor geodesic momentum. Bold indicates the lower value within each pair.

| Model | Steps | Baseline PPL  | Geodesic PPL   | Gain <sub>PPL</sub> | Baseline JS    | Geodesic JS    |
|-------|-------|---------------|----------------|---------------------|----------------|----------------|
| MDLM  | 10    | 551.953       | <b>407.473</b> | +26.18%             | <b>0.38659</b> | 0.38850        |
|       | 20    | 261.453       | <b>212.993</b> | +18.53%             | <b>0.37718</b> | 0.38073        |
|       | 50    | 140.075       | <b>120.542</b> | +13.95%             | <b>0.36864</b> | 0.37629        |
|       | 100   | 92.421        | <b>81.864</b>  | +11.42%             | 0.38260        | <b>0.38185</b> |
|       | 256   | <b>65.431</b> | 65.665         | −0.36%              | <b>0.37828</b> | 0.38066        |
|       | 512   | <b>50.905</b> | 51.801         | −1.76%              | 0.38224        | <b>0.37800</b> |
| SEDD  | 10    | 486.276       | <b>434.932</b> | +10.56%             | <b>0.35628</b> | 0.36043        |
|       | 20    | 235.867       | <b>214.863</b> | +8.91%              | 0.34810        | <b>0.34707</b> |
|       | 50    | 123.169       | <b>114.949</b> | +6.67%              | 0.34771        | <b>0.34503</b> |
|       | 100   | <b>82.088</b> | 84.184         | −2.55%              | 0.35678        | <b>0.35078</b> |
|       | 256   | <b>59.954</b> | 63.276         | −5.54%              | <b>0.34519</b> | 0.34670        |
|       | 512   | 51.695        | <b>50.903</b>  | +1.53%              | <b>0.36408</b> | 0.36661        |

20, 50, 100, 256, and 512. The quality metrics are:

- token-weighted perplexity under GPT-2, where lower is better; and
- Jensen–Shannon divergence between generated and reference unigram token histograms, where lower is better. References are nonempty strings from the WikiText-2 raw validation subset.

## 5.2 Quality results

At 10, 20, 50, and 100 MDLM steps, the geodesic samples have respectively 26.18%, 18.53%, 13.95%, and 11.42% lower GPT-2 perplexity. At 256 and 512 steps, the difference vanishes and slightly favors the baseline. This pattern is consistent with the method’s intended use as a low-step correction. The MDLM damping in [equation \(16\)](#) also decreases  $\alpha$  beyond 50 uniform steps: approximately 0.25 at 100, 0.098 at 256, and 0.049 at 512.

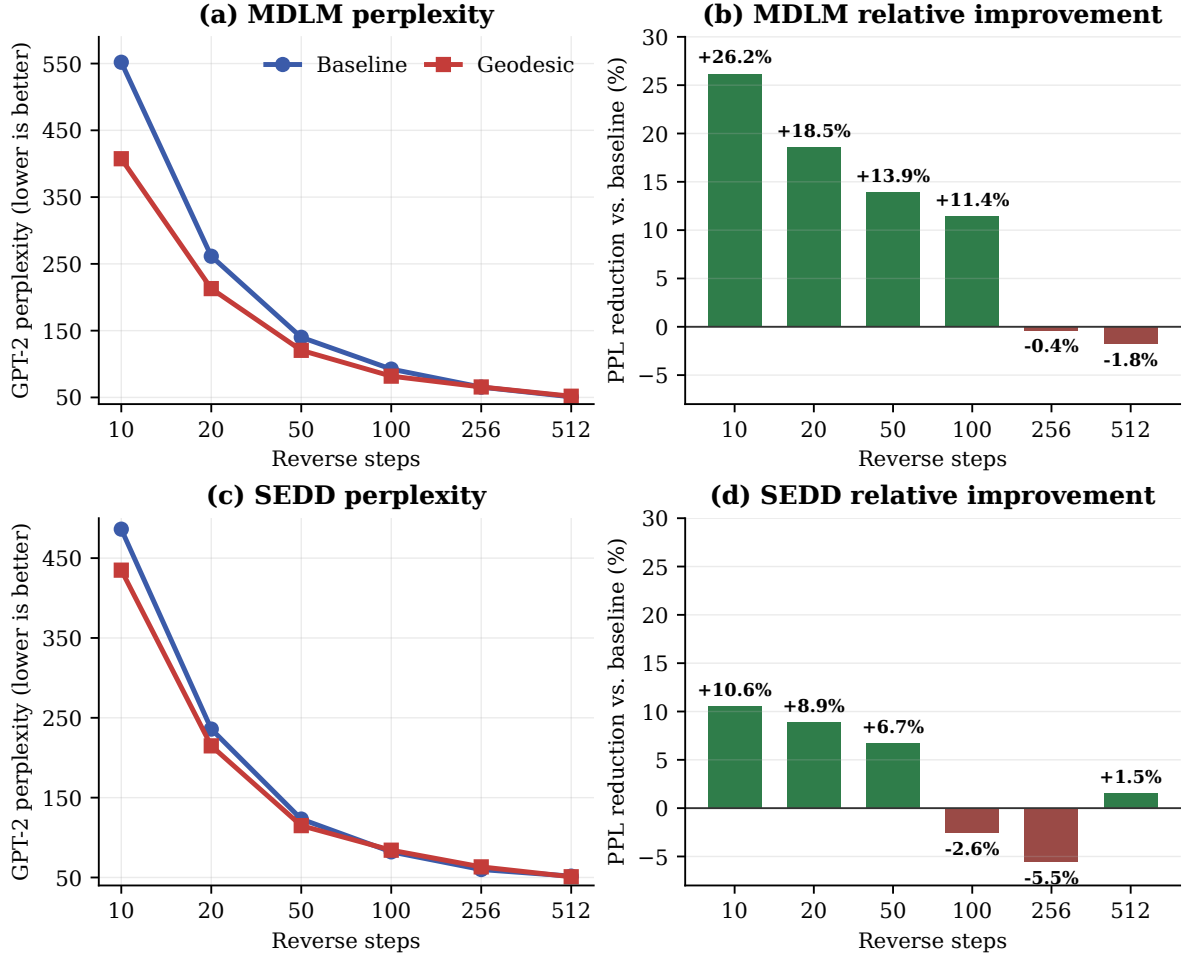
SEDD shows smaller perplexity reductions at 10–50 steps, degradation at 100–256, and a small improvement at 512. Its implementation used  $\alpha = 1/2$  after the first step on every uniform grid, without MDLM’s high-step damping. This may contribute to overshoot, but the available experiment does not isolate the cause. Jensen–Shannon divergence is mixed for both models and does not show a consistent distributional improvement.

## 6 Discussion

**Interpretation.** The strongest defensible interpretation is prediction-lag correction. With few reverse steps, successive denoiser or transition distributions can move substantially. Continuing their recent Fisher–Rao arc may anticipate the next prediction and improve quality. As the grid is refined, raw predictions change less, so extrapolation offers little benefit and can amplify stochastic or model error.

**MDLM versus SEDD.** MDLM applies momentum to a clean-token prediction before the absorbing transition and freezes completed tokens. SEDD applies momentum to a stochastic-state-dependent one-step transition. The latter object may be less temporally smooth, which weakens the curve-extrapolation interpretation and is consistent with the less stable archived trend.

**Relationship to continuous geometric models.** The method uses the same Fisher–Rao embedding as Fisher-Flow and RDLM but not their dynamics. A genuine Riemannian multistep solver would require a continuous state  $X_t$ , a well-defined tangent vector field, parallel transport between  $T_{X_{t-1}}\mathcal{M}$  and  $T_{X_t}\mathcal{M}$ , and



**Figure 1:** Perplexity results. Left panels show raw GPT-2 perplexity on linear vertical axes fitted separately to each model’s observed range, making the low-step gaps directly visible. Right panels show the relative PPL reduction  $(\text{baseline} - \text{geodesic})/\text{baseline}$ , so positive bars indicate improvement and negative bars indicate degradation. Both improvement panels retain the same  $[-8, 30]$  percentage-point range.

an exponential-map state update. Establishing such a solver for pretrained discrete MDLM or SEDD would require a new derivation connecting their categorical predictions to a continuous vector field.

## 7 Limitations

1. **No convergence order.** Stochastic jumps, point rather than vector extrapolation, recursive history, and nonsmooth projection invalidate a direct AB2 order argument.
2. **Boundary geometry.** Exact zeros place predictions on the boundary where the Fisher–Rao metric is singular.
3. **Position-wise approximation.** Each token position is treated as a separate categorical simplex. The geometry of the full joint sequence distribution is not represented.

## 8 Conclusion

We presented geodesic momentum sampling, an inference-time modification for MDLM and SEDD that recursively extrapolates categorical predictions on the Fisher–Rao square-root sphere. The update has an exact exponential-map interpretation before positive-orthant projection, and its local coefficients match variable-step AB2 coefficients. The algorithm nevertheless operates on prediction points along a stochastic discrete path, not on parallel-transported vector fields of a probability-flow ODE. Its mathematically accurate characterization is therefore *recursive Fisher–Rao spherical extrapolation*, not a proven second-order Riemannian solver.

Results show promising low-step perplexity behavior, especially for MDLM, while distributional metrics and higher-step SEDD behavior are mixed. A derivation based on a continuous vector field remains necessary for formal convergence claims.

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